Econ 101A
Section 13

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Last section we found prices and quantities. Today we ask whether the outcome is any good. The yardstick is surplus: what buyers would have paid on top of what they pay, and what sellers receive on top of their costs. With it we can put a number on the cost of a tax, the gains from trade, and the damage from rent control. Then we let firms enter, which pushes profits to zero. We end with a first look at monopoly. This is the second half of Lecture 9 and the start of Lecture 10.

We reuse one market all day: 10 firms with cost $c(y) = 4 + y^2$ (the 4 is avoided by shutting down), so $Y^S = 5p$ for $p \geq 4$ and 0 below; and 5 consumers, so $X^D = 100 - 5p$. Without a tax, $p^* = 10$, $Q^* = 50$, and each firm earns 21.

1 Producer Surplus

Producer surplus is profit, $\pi(p, y_0) = p\, y_0 - c(y_0)$. Two pictures:

Rectangle. $\pi = y_0 \big(p - \mathrm{AC}(y_0)\big)$: quantity times the margin over average cost.

Area above MC. Since $c(y_0) = c(0) + \int_0^{y_0} c'(y)\, dy$,

$$\pi = \int_0^{y_0} \big(p - c'(y)\big)\, dy - c(0).$$

Profit is the area between the price and marginal cost, minus the fixed cost.

A third picture is often the easiest. By the envelope theorem $d\pi/dp = y(p)$, and profit is zero at the shutdown price $\underline{p}$. So $\pi(p) = \int_{\underline{p}}^{p} y(s)\, ds$: the area to the left of the supply curve, between $\underline{p}$ and p.

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2 Consumer Surplus

When the price goes from p_0 to p_1, how much money would make the consumer as well off as before? That is $e(p_1, u) - e(p_0, u)$. By the envelope theorem $\partial e/\partial p = h(p, u)$, the Hicksian demand. So

$$e(p_1, u) - e(p_0, u) = \int_{p_0}^{p_1} h(p, u)\, dp,$$

the area to the left of the *Hicksian* demand curve. In practice we use the area under the ordinary (Marshallian) demand curve. The two coincide when there are no income effects, and are close when the good is a small share of the budget.

3 Total Surplus and Deadweight Loss

Total surplus is consumer surplus plus producer surplus plus any tax revenue. In a competitive market without distortions it is as large as it can be. A tax, a tariff, or a price ceiling makes some trades that are worth more than they cost not happen. The surplus lost on those trades is the deadweight loss.

4 The Long Run

In the long run firms can enter. They enter as long as profit is positive, that is, as long as $p > \text{AC}$. Entry shifts industry supply right and pushes the price down until $p = \min \text{AC}$. At that point $\text{MC} = \text{AC}$ and each firm earns zero. If each firm's cost does not depend on how many firms there are (a *constant cost industry*), the long-run supply curve is flat at $\min \text{AC}$.

5 Exercises

Exercise 1 (producer surplus three ways) A firm has $c(y) = 4 + y^2$ for $y > 0$, $c(0) = 0$. The price is $p = 10$.

(a) Find output and profit.

(b) Compute profit as a rectangle, $y_0\,(p - \text{AC})$.

(c) Compute profit as the area between p and MC, minus the setup cost.

(d) Compute profit as the area to the left of the supply curve, between the shutdown price and p. Why does this work? Use it to get industry producer surplus with 10 firms.

Exercise 2 (when is the area under demand the right measure?)

(a) A consumer has quasilinear utility $u(x, m) = 20x - \frac{1}{2}x^2 + m$, where m is money spent on everything else. Find the Marshallian demand, the Hicksian demand and the expenditure function. Check that $\partial e/\partial p = h$. Find the welfare loss when p rises from 10 to 12, and the consumer surplus at $p = 10$.

(b) A consumer has $u(x, y) = x^{1/2}y^{1/2}$, income $M = 100$, $p_y = 1$. The price of x rises from 1 to 4. Find the Hicksian demand and expenditure function. How much money would the consumer need to be as well off as before? Compare with the area under the Marshallian demand curve.

Exercise 3 (the cost of a tax) Use the market above. Consumers have the quasilinear utility of Exercise 2(a), so the area under demand is exact.

(a) Compute consumer surplus, producer surplus and total surplus without a tax.

(b) A tax of $t = 4$ per unit is levied. From Section 12, buyers pay 12, sellers keep 8, $Q = 40$. Compute consumer surplus, producer surplus, tax revenue, total surplus and the deadweight loss. Draw the picture.

(c) Check the deadweight loss with a triangle. Who loses what?

(d) Double the tax to $t = 8$. What happens to revenue and to the deadweight loss?

(e) Suppose instead supply were vertical. What would the deadweight loss of a tax be?

Exercise 4 (opening to trade) The same domestic market (10 firms, short run) opens to trade. The good can be bought on the world market at price p_T, in any quantity.

(a) Let $p_T = 6$. Find domestic production, consumption and imports. Compute CS, PS and total surplus. Who wins, who loses, and what is the net gain? Draw the picture.

(b) The government adds a tariff of 2 per imported unit. Redo (a). Where is the deadweight loss?

(c) Let $p_T = 3$ instead. What happens to the domestic industry?

Exercise 5 (rent control) In a rental market, demand is $Q^D = 120 - 2p$ and supply is $Q^S = 2p - 40$. The city caps rent at $\bar{p} = 30$.

(a) Find the equilibrium without the cap, and CS, PS.

(b) With the cap, how many units are rented? How many people want one?

(c) *Cost 1.* Assume the units go to the people who value them most. Compute CS, PS and the deadweight loss. Do tenants gain?

(d) *Cost 2.* Now assume the units go at random to people willing to pay at least 30. Compute CS and the total loss. Do tenants as a group still gain?

(e) In the long run, landlords can convert rental units into condos. What does that do to the cost of rent control?

Exercise 6 (free entry) Keep the firm $c(y) = 4 + y^2$ and demand $X^D = 100 - 5p$, but now any number of identical firms can enter.

(a) Find the long-run price, output per firm, total quantity and number of firms.

(b) Demand rises to $X^D = 140 - 5p$. Find the short-run equilibrium with the number of firms from (a), then the new long-run equilibrium.

(c) Back to $X^D = 100 - 5p$. A tax of $t = 2$ per unit is levied. Find the long-run outcome. Who pays the tax? Compare with Section 12, Exercise 5.

Exercise 7 (preview: monopoly) Now one firm serves the whole market. It produces at constant marginal cost 4 with no fixed cost. Inverse demand is $p = 20 - Q/5$.

(a) Find the monopoly quantity, price and profit.

(b) Check the markup rule $\dfrac{p - \mathrm{MC}}{p} = -\dfrac{1}{\varepsilon_{Q,p}}$.

(c) Compare with the competitive outcome. What is the deadweight loss?

(d) Compare with a tax of $t = 8$ in the competitive industry of Exercise 6. What do you notice?